1. Clinical Question & Core Hypothesis

Clinical Question

TLS/high in-situ lymphoid organization is generally expected to improve anti-tumor immunity and prognosis, yet many immune-infiltrated localized high■risk tumors still have poor overall survival—why does local immune organization sometimes fail to translate into clinical benefit? 2) Tier 1 paths used: I built on CORE PATH 1 (status of in■situ immune cell infiltrate → imaging/pathomics readouts of immune organization) and CORE PATH 2 (PTGER4 → PGE2-driven immunosuppression/T■cell dysfunction) as the primary mechanistic axis; I included CORE PATH 3 (SOX2/NE features) as a second tumor■intrinsic modifier because NE programs plausibly blunt immune efficacy and are linked to poor outcomes. 3) Tier 2 evidence supporting the claim: Decipher GC and CENPA offer orthogonal genomic risk stratification that can help identify high■risk TLS■rich tumors; inflammation signatures (EDIH) and urinary biomarkers can act as systemic or noninvasive readouts of the immunometabolic state that may correlate with PTGER4 activity and NE transdifferentiation. 4) Why OS as endpoint: the top paths and clinical signals converge on survival (OS/metastasis■free survival) as the most robust, clinically meaningful outcome for localized high■risk patients and is directly represented in the path rankings. 5) Deliberate exclusions: I excluded exploratory Tier 3 signals (AFP, insulin■response markers, CSF■1/SPC) from the core hypothesis to maintain mechanistic focus; I also limited anchor genes to PTGER4 and SOX2 (excluded naming other immune checkpoint genes in the hypothesis) to satisfy the 1–2 anchor constraint and sharpen a testable mechanistic question?

Working Hypothesis

Among TLS-rich localized high-risk tumors, tumor-cell expression of PTGER4 and concurrent SOX2-driven neuroendocrine programs define biologically distinct subgroups with divergent overall survival (OS). Specifically, high PTGER4 signaling (PGE2–PTGER4 axis) in TLS-rich tumors promotes local T-cell dysfunction and functional impairment within TLS, attenuating TLS-mediated anti-tumor immunity. This effect is amplified by SOX2-associated neuroendocrine programs that reprogram the tumor immune microenvironment by reducing antigen presentation and suppressing TLS-supportive chemokine signaling. Consequently, TLS-high, PTGER4-high, and SOX2-high tumors will exhibit the poorest OS, whereas TLS-high tumors that are PTGER4-low and/or SOX2-low will retain TLS-associated survival benefits. Secondary modifiers (Decipher_GC, CENPA, EDIH, and urinary biomarkers) will correlate with PTGER4 activity and NE state, improving prognostic stratification within TLS-rich cohorts. This framework yields testable predictions: (1) within TLS-high tumors, PTGER4 expression and SOX2 co-segregate with worse OS; (2) spatial proximity of PTGER4-high tumor cells to TLS regions correlates with reduced T-cell activation/exhaustion marker patterns locally; (3) functional assays will show PTGER4 dependency for PGE2-mediated T-cell suppression, and (4) multi-omic signatures will associate with outcome across independent cohorts.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PTGER4, SOX2
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PTGER4: PTGER4 → PGE2-driven immunosuppression/T■cell dysfunction) as the primary mechanistic axis; I included CORE PATH 3 (SOX2/NE features) as a second tumor■intrinsic modifier because NE programs plausibly blunt immune efficacy and are linked to poor outcomes.

SOX2: SOX2 (excluded naming other immune checkpoint genes in the hypothesis) to satisfy the 1–2 anchor constraint and sharpen a testable mechanistic question.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Decipher_GC	Supporting modifier from knowledge graph
Secondary	CENPA	Supporting modifier from knowledge graph
Secondary	EDIH	Supporting modifier from knowledge graph
Secondary	urinary_biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: provide a focused, testable core, while tier-2 modifiers (Decipher_GC, CENPA, EDIH, urinary biomarkers) offer orthogonal readouts of genomic risk and systemic/inflammatory state that may modulate the TLS–immunity–survival relationship.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Decipher_GC: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CENPA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.